

Data Analysis Final Project (G2)

Business Context

Business Context

AdventureWorks is a fictional multinational company that manufactures and sells bicycles, bicycle components, and accessories through both individual customers and reseller stores. The company records large volumes of operational data from sales, products, customers, shipping, and geographic locations. Although this transactional data supports daily business operations, it is not optimized for analytical reporting or business intelligence.

Business Objectives

The main objectives of this project are:

- Build a centralized Data Warehouse from the AdventureWorks operational database.
- Improve data quality through data profiling and cleansing.
- Integrate data from multiple business entities into a consistent analytical model.
- Design a Star Schema to simplify analytical queries.
- Support business intelligence reporting using Power BI.
- Enable faster and more reliable business decision-making.

Data Source

- **Data Source**
- The data used in this project comes from the **AdventureWorks2022** sample database, which is a Microsoft SQL Server Online Transaction Processing (OLTP) database. The database simulates the daily operations of a manufacturing and retail company and contains detailed information about customers, products, orders, sales, shipping, and geographic locations.

Introduction

Before starting the ETL transformation process, the Bronze layer was profiled to assess the quality of the source data. Several validation checks were performed to identify potential issues such as duplicates, missing values, invalid data, inconsistent categories, and broken relationships. The results of these checks were used to determine the appropriate cleaning actions before moving the data to the Silver layer.

Test 1: Duplicate Primary Keys

Objective

Verify that primary keys are unique.

SQL Used

```
9  SELECT 'Product' AS [Table], 'Duplicate ProductID' AS [Anomaly_Type],
10     COUNT(*) AS [Affected_Row_Count]
11  FROM (SELECT ProductID FROM bronze.[Product] GROUP BY ProductID HAVING COUNT(*) > 1) d;
12
13  SELECT 'Customer' AS [Table], 'Duplicate CustomerID' AS [Anomaly_Type],
14     COUNT(*) AS [Affected_Row_Count]
15  FROM (SELECT CustomerID FROM bronze.Customer GROUP BY CustomerID HAVING COUNT(*) > 1) d;
16
17  SELECT 'SalesOrderHeader' AS [Table], 'Duplicate SalesOrderID' AS [Anomaly_Type],
18     COUNT(*) AS [Affected_Row_Count]
19  FROM (SELECT SalesOrderID FROM bronze.SalesOrderHeader GROUP BY SalesOrderID HAVING COUNT(*) > 1) d;
20  GO
```

Result

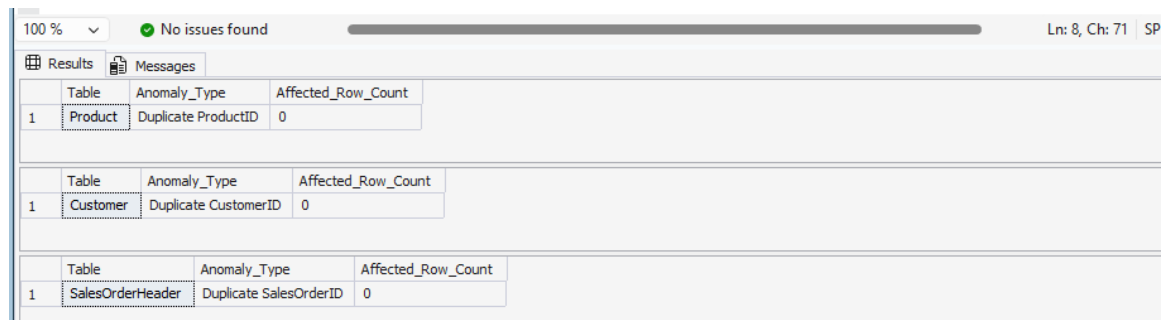


Table	Anomaly_Type	Affected_Row_Count
1 Product	Duplicate ProductID	0

Table	Anomaly_Type	Affected_Row_Count
1 Customer	Duplicate CustomerID	0

Table	Anomaly_Type	Affected_Row_Count
1 SalesOrderHeader	Duplicate SalesOrderID	0

- Duplicate ProductID = 0
- Duplicate CustomerID = 0
- Duplicate SalesOrderID = 0

Decision

No action was required because all primary keys are unique.

Test 2: Numeric Value Validation

Objective

Validate business rules for numeric fields.

SQL Used

```
25 SELECT 'SalesOrderDetail' AS [Table], 'Non-positive OrderQty' AS [Anomaly_Type],
26        COUNT(*) AS [Affected_Row_Count], MIN(OrderQty) AS [Example_Messy_Value]
27 FROM bronze.SalesOrderDetail
28 WHERE OrderQty <= 0;
29
30 SELECT 'SalesOrderDetail' AS [Table], 'Negative UnitPrice' AS [Anomaly_Type],
31        COUNT(*) AS [Affected_Row_Count], MIN(UnitPrice) AS [Example_Messy_Value]
32 FROM bronze.SalesOrderDetail
33 WHERE UnitPrice < 0;
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```

All checks returned **0 invalid records**.

Note: Whenever Example_Messy_Value appears as **NULL**, it does **not** indicate missing data. It simply means that no records violated the validation rule, so there was no invalid value to display.

Decision

No cleaning was required because all numeric values satisfy the business rules.

Test 3: Date Validation

Objective

Check that business dates follow the correct sequence.

SQL Used

```
50 SELECT 'SalesOrderHeader' AS [Table], 'ShipDate before OrderDate' AS [Anomaly_Type],
51        COUNT(*) AS [Affected_Row_Count]
52 FROM bronze.SalesOrderHeader
53 WHERE ShipDate < OrderDate;
54
55 SELECT 'SalesOrderHeader' AS [Table], 'DueDate before OrderDate' AS [Anomaly_Type],
56        COUNT(*) AS [Affected_Row_Count]
57 FROM bronze.SalesOrderHeader
58 WHERE DueDate < OrderDate;
59 GO
60
```

Result

100 %
No issues found
Ln: 53, Ch: 28 SPC CRLF UTF-8

Table	Anomaly_Type	Affected_Row_Count
1 SalesOrderHeader	ShipDate before OrderDate	0

Table	Anomaly_Type	Affected_Row_Count
1 SalesOrderHeader	DueDate before OrderDate	0

Both checks returned **0 records**.

Decision

No action was required because all dates are logically consistent.

Test 4: Missing Values Inspection

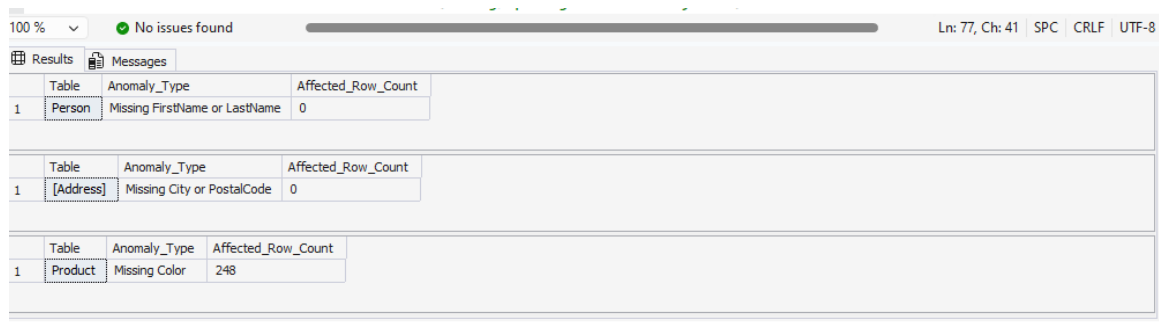
Objective

Identify missing values in important columns.

SQL Used

```
68
69 SELECT '[Address]' AS [Table], 'Missing City or PostalCode' AS [Anomaly_Type],
70        COUNT(*) AS [Affected_Row_Count]
71 FROM bronze.[Address]
72 WHERE City IS NULL OR TRIM(City) = '' OR PostalCode IS NULL OR TRIM(PostalCode) = '';
73
74 SELECT '[Product]' AS [Table], 'Missing Color' AS [Anomaly_Type],
75        COUNT(*) AS [Affected_Row_Count]
76 FROM bronze.[Product]
77 WHERE Color IS NULL OR TRIM(Color) = '';]
78 GO
--
```

Result



The screenshot shows the SQL Server Enterprise Manager interface. At the top, there's a status bar indicating '100 %' zoom, 'No issues found', and file encoding settings (Ln: 77, Ch: 41, SPC, CRLF, UTF-8). Below this, the 'Results' pane displays three tables of data. Each table has three columns: 'Table', 'Anomaly_Type', and 'Affected_Row_Count'. The first table shows 0 missing first or last names for the 'Person' table. The second table shows 0 missing city or postal codes for the 'Address' table. The third table shows 248 missing colors for the 'Product' table.

	Table	Anomaly_Type	Affected_Row_Count
1	Person	Missing FirstName or LastName	0

	Table	Anomaly_Type	Affected_Row_Count
1	[Address]	Missing City or PostalCode	0

	Table	Anomaly_Type	Affected_Row_Count
1	Product	Missing Color	248

- Missing FirstName/LastName = **0**
- Missing City/PostalCode = **0**
- Missing Color = **248**

Additional Investigation

To understand why Color was missing:

```

153 SELECT
154     pc.Name AS Category,
155     COUNT(*) AS MissingColorCount
156 FROM bronze.Product p
157 LEFT JOIN bronze.ProductSubcategory ps
158     ON p.ProductSubcategoryID = ps.ProductSubcategoryID
159 LEFT JOIN bronze.ProductCategory pc
160     ON ps.ProductCategoryID = pc.ProductCategoryID
161 WHERE p.Color IS NULL
162        OR TRIM(p.Color) = ''
163 GROUP BY pc.Name
164 ORDER BY MissingColorCount DESC;
165

```

100 % No issues found Ln: 162, Ch: 25 SPC CRLF UTF-8

	Category	MissingColorCount
1	NULL	198
2	Components	26
3	Accessories	24

identifying the categories of products with missing colors

```

197 SELECT ProductID,
198         Name,
199         ProductNumber,
200         ProductSubcategoryID
201 FROM bronze.Product
202 WHERE Color IS NULL;
203

```

100 % No issues found Ln: 203, Ch: 21 SPC CRLF UTF-8

	ProductID	Name	ProductNumber	ProductSubcategoryID
1	1	Adjustable Race	AR-5381	NULL
2	2	Bearing Ball	BA-8327	NULL
3	3	BB Ball Bearing	BE-2349	NULL
4	4	Headset Ball Bearings	BE-2908	NULL
5	316	Blade	BL-2036	NULL
6	323	Crown Race	CR-9981	NULL
7	324	Chain Stays	CS-2812	NULL
8	325	Decal 1	DC-8732	NULL
9	326	Decal 2	DC-9824	NULL

Query executed successfully. (local) (17.0 RTM) FIREFOX\Firefox (95) AdventureWorks_DW 00:00:00 Row: 1, Col: 1 248 rows

Understanding which products have missing colors

Result

Products such as **Bearing Ball, Chainring, Flat Washer, Crankarm, Fork End**, and other bicycle components were identified.

```

175 SELECT
176     COUNT(*) AS ProductsWithoutSubcategory
177 FROM bronze.Product
178 WHERE ProductSubcategoryID IS NULL;
179
180

```

100 % No issues found Ln: 174, Ch: 1 SPC CRLF UTF-8

	ProductsWithoutSubcategory
1	209

verifying whether products without a subcategory also have missing colors



```
181 SELECT
182     ProductID,
183     Name,
184     Color
185 FROM bronze.Product
186 WHERE ProductSubcategoryID IS NULL
187     AND Color IS NOT NULL;
```

	ProductID	Name	Color
1	317	LL Crankarm	Black
2	318	ML Crankarm	Black
3	319	HL Crankarm	Black
4	320	Chainring Bolts	Silver
5	321	Chainring Nut	Silver
6	322	Chainring	Black
7	332	Freewheel	Silver
8	351	Front Derailleur Cage	Silver
9	352	Front Derailleur Linkage	Silver

Query executed successfully. (local) (17.0 RTM) FIREFOX\Firefox (95) AdventureWorks_DW | 00:00:00 | Row: 1, Col: 1 | 11 rows

Identifying products without a subcategory but with an assigned color

Result

- **11 products** have a valid color (Black or Silver), including products such as **LL Crankarm**, **Chainring**, **Freewheel**, and **Lock Ring**.

Decision

The profiling identified **248 products with missing Color values**. Further investigation showed that most of these products are bicycle components or products without an assigned subcategory. Since **Color is an optional descriptive attribute**, the missing values do not affect business processes or analytical calculations. Therefore, the records were retained, and the missing values can be replaced with **"Unknown"** in the Silver layer for reporting purposes

Test 5: Categorical Value Inspection

Objective

Check for inconsistent text values caused by different casing or extra spaces.

SQL Used

```
80  -- =====
81  -- TEST 5: CATEGORICAL VALUE INSPECTION (casing/spacing inconsistency risk)
82  -- =====
83  SELECT 'Product' AS [Table], 'Distinct row Color values' AS [Anomaly_Type],
84         COUNT(DISTINCT Color) AS [Distinct_Raw_Count]
85  FROM bronze.[Product];
86
87  SELECT 'CountryRegion' AS [Table], 'Distinct row Name values' AS [Anomaly_Type],
88         COUNT(DISTINCT Name) AS [Distinct_Raw_Count]
89  FROM bronze.CountryRegion;
90  GO
91
```

00 % | No issues found | Ln: 89, Ch: 27 | SPC | CRLF | UTF-8

Results | Messages

	Table	Anomaly_Type	Distinct_Raw_Count
1	Product	Distinct row Color values	9

	Table	Anomaly_Type	Distinct_Raw_Count
1	CountryRegion	Distinct row Name values	238

```
193  SELECT DISTINCT Color
194  FROM bronze.Product
195  ORDER BY Color;
196
197  SELECT DISTINCT Name
198  FROM bronze.CountryRegion
199  ORDER BY Name;
```

100 % | No issues found | Ln: 200, Ch: 15 | SPC | CRLF | UTF-8

Results | Messages

	Color
1	NULL
2	Black
3	Blue
4	Grey
5	Multi

	Name
1	Afghanistan
2	Albania
3	Algeria
4	American ...

Query executed successfully. | (local) (17.0 RTM) | FIREFOX\Firefox (95) | AdventureWorks_DW | 00:00:00 | Row: 1, Col: 1 | 248 rows

Result:

Result: SELECT DISTINCT confirmed that Product Color contains **9 consistent values** and CountryRegion contains **238 unique names**. No inconsistencies related to casing, spacing, or spelling were found. Therefore, no standardization was required.

Decision

No standardization was required.

Test 6: Referential Integrity Validation

Objective

Verify that all foreign keys reference existing parent records.

SQL Used

```
96 -- 6a. Sales order lines pointing to non-existent Products
97 SELECT 'SalesOrderDetail' AS [Table], 'Orphaned ProductID' AS [Anomaly_Type],
98        COUNT(*) AS [Affected_Row_Count]
99 FROM bronze.SalesOrderDetail d
100 LEFT JOIN bronze.[Product] p ON d.ProductID = p.ProductID
101 WHERE p.ProductID IS NULL;
102
103 -- 6b. Sales orders pointing to non-existent Customers
104 SELECT 'SalesOrderHeader' AS [Table], 'Orphaned CustomerID' AS [Anomaly_Type],
105        COUNT(*) AS [Affected_Row_Count]
106 FROM bronze.SalesOrderHeader h
107 LEFT JOIN bronze.Customer c ON h.CustomerID = c.CustomerID
108 WHERE c.CustomerID IS NULL;
109
```

Result

Table	Anomaly_Type	Affected_Row_Count
SalesOrderDetail	Orphaned ProductID	0
SalesOrderHeader	Orphaned CustomerID	0
Product	Orphaned ProductSubcategoryID	0
ProductSubcategory	Orphaned ProductCategoryID	0
ProductCategory	Orphaned ProductCategoryID	0

All referential integrity checks returned **0 orphan records**.

Decision

No action was required because all relationships are valid.

Profiling Conclusion

The data profiling process confirmed that the Bronze dataset is largely clean and consistent. Most validation tests passed without identifying any data quality issues. The only finding was missing values in the **Color** attribute, which were determined to be non-critical after further investigation. Since Color is an optional descriptive field, the records were retained, and the missing values may be handled in the Silver layer if needed. Overall, the dataset is ready for the next stage of the ETL pipeline.

Data Dictionary -- Silver Layer

silver.ProductCategory

Top-level product classification (e.g. Bikes, Components, Clothing, Accessories).

Column	Description
ProductCategoryID	Primary key, unique category identifier
Name	Category name, standardized to uppercase

silver.ProductSubcategory

Second-level product classification, nested under a category.

Column	Description
ProductSubcategoryID	Primary key, unique subcategory identifier
ProductCategoryID	Foreign key reference to the parent category
Name	Subcategory name, standardized to uppercase

silver.Product

Cleaned catalog of all AdventureWorks products.

Column	Description
ProductID	Primary key, unique product identifier
Name	Product display name
ProductNumber	Manufacturer/internal product code
Color	Product color; blank values standardized to 'Unknown'
Size	Product size code
Weight	Product weight
ProductLine	Product line code
Class	Product class code
Style	Product style code
StandardCost	Standard manufacturing cost; guarded against negative values
ListPrice	Retail list price; guarded against negative/null values
ProductSubcategoryID	Foreign key reference to the subcategory

silver.SalesTerritory

Sales regions used to attribute revenue geographically.

Column	Description
TerritoryID	Primary key, unique territory identifier
Name	Territory display name

Column	Description
CountryRegionCode	Country code the territory belongs to
Group	Higher-level regional grouping (e.g. North America, Europe)
SalesYTD	Year-to-date sales for the territory
SalesLastYear	Prior-year sales for the territory
CostYTD	Year-to-date cost for the territory
CostLastYear	Prior-year cost for the territory

silver.CountryRegion

Reference list of countries/regions.

Column	Description
CountryRegionCode	Primary key, ISO-style country code
Name	Country/region display name

silver.StateProvince

State/province reference data, linked to both country and sales territory.

Column	Description
StateProvinceID	Primary key, unique state/province identifier
StateProvinceCode	Short state/province code
CountryRegionCode	Foreign key reference to the country/region
Name	State/province display name

Column	Description
TerritoryID	Foreign key reference to the sales territory

silver.Address

Cleaned physical address records used for billing and shipping.

Column	Description
AddressID	Primary key, unique address identifier
AddressLine1	Primary street address line
AddressLine2	Secondary street address line (optional)
City	City name
StateProvinceID	Foreign key reference to the state/province
PostalCode	Postal/ZIP code

silver.Person

Individual contact records (used for individual customers).

Column	Description
BusinessEntityID	Primary key, unique person identifier
PersonType	Short code identifying the type of person record
Title	Honorific title (Mr., Ms., etc.)
FirstName	First name
MiddleName	Middle name (optional)

Column	Description
LastName	Last name
Suffix	Name suffix (Jr., Sr., etc.), if any
EmailPromotion	Email promotion opt-in level
AdditionalContactInfo	Supplementary contact info, stored as plain XML
Demographics	Demographic survey data, stored as plain XML

silver.Store

Retail store records (used for reseller/store customers).

Column	Description
BusinessEntityID	Primary key, unique store identifier
Name	Store display name
SalesPersonID	Sales representative assigned to the store
Demographics	Store demographic/survey data, stored as plain XML

silver.Customer

Bridge entity linking either a Person or a Store to sales activity.

Column	Description
CustomerID	Primary key, unique customer identifier
PersonID	Foreign key reference to silver.Person (individual customers)
StoreID	Foreign key reference to silver.Store (reseller customers)

Column	Description
TerritoryID	Foreign key reference to the sales territory
AccountNumber	Customer account number

silver.SalesOrderHeader

One row per sales order (order-level information).

Column	Description
SalesOrderID	Primary key, unique order identifier
RevisionNumber	Order revision counter
OrderDate	Date the order was placed
DueDate	Date the order is due
ShipDate	Date the order was shipped
Status	Order status code
OnlineOrderFlag	1 if placed online, 0 if placed through a sales rep/store
SalesOrderNumber	Human-readable order number
PurchaseOrderNumber	Customer's purchase order reference
AccountNumber	Customer account number on the order
CustomerID	Foreign key reference to the customer
SalesPersonID	Sales representative who processed the order
TerritoryID	Sales territory the order is attributed to
BillToAddressID	Foreign key reference to the billing address

Column	Description
ShipToAddressID	Foreign key reference to the shipping address
ShipMethodID	Foreign key reference to the shipping method
CreditCardID	Foreign key reference to the credit card used
CreditCardApprovalCode	Payment approval code
CurrencyRateID	Currency conversion rate applied
SubTotal	Order subtotal before tax/freight
TaxAmt	Tax amount
Freight	Freight/shipping charge
TotalDue	Total amount due
Comment	Free-text order comment

silver.SalesOrderDetail

One row per line item within a sales order.

Column	Description
SalesOrderDetailID	Primary key, unique line item identifier
SalesOrderID	Foreign key reference to the parent order
CarrierTrackingNumber	Shipping carrier tracking number
OrderQty	Quantity ordered; guarded to be strictly positive
ProductID	Foreign key reference to the product
SpecialOfferID	Promotional offer applied to the line

Column	Description
UnitPrice	Price per unit; guarded to be non-negative
UnitPriceDiscount	Discount rate applied to the unit price
LineTotal	Total value of the line item

silver.ShipMethod

Reference list of shipping methods and their base rates.

Column	Description
ShipMethodID	Primary key, unique shipping method identifier
Name	Shipping method display name
ShipBase	Base shipping charge
ShipRate	Per-unit shipping rate

silver.CreditCard

Reference list of credit cards used for payment.

Column	Description
CreditCardID	Primary key, unique credit card record identifier
CardType	Card brand/type (Visa, MasterCard, etc.)
CardNumber	Card number (sensitive data -- consider masking in reporting)
ExpMonth	Expiration month
ExpYear	Expiration year

Data Dictionary -- Gold Layer

1. gold.FactSales (Fact Table)

Column	Description
SalesDetailKey	Unique identifier for each line item within a sales order.
OrderKey	Foreign Key referencing the specific sales order header identifier.
OrderDateKey	Foreign Key referencing DimDate for order placement date (YYYYMMDD).
DueDateKey	Foreign Key referencing DimDate for payment due date (YYYYMMDD).
ShipDateKey	Foreign Key referencing DimDate for shipping date (YYYYMMDD).
SK_Customer	Surrogate Key referencing the customer in DimCustomer.
SK_Product	Surrogate Key referencing the purchased product in DimProduct.
ShipGeographyKey	Surrogate Key referencing shipping destination in DimGeography.
SK_ShipMethod	Surrogate Key referencing shipping method in DimShipMethod.
SK_CreditCard	Surrogate Key referencing credit card details in DimCreditCard.
SalesChannel	Indicator label denoting order origin ('Online' or 'Reseller / Store').
OrderQuantity	Number of units ordered for the specific line item.
UnitPrice	Selling price per unit of product.
UnitPriceDiscount	Discount amount applied per unit.

2. gold.DimProduct

Column	Description
SK_Product	Primary Surrogate Key representing each individual product version.
ProductID	Original business identifier for the product.
ProductName	Official commercial name of the product.
ProductNumber	Unique alphanumeric product code.
ProductColor	Color attribute of the product.
Size	Surrogate Key referencing the customer in DimCustomer.
Weight	Weight attribute of the product
ProductLine	High-level product line category flag.
Style	Design style category.
StandardCost	Standard manufacturing or acquiring cost of the product
ListPrice	Recommended retail listing price.
SubcategoryName	Top-level parent category name for the product
StartDate	Effective start date for the record version (SCD Type 2).
EndDate	Expiration date for historical record versions (SCD Type 2).
IsCurrent	Active status flag (1 = Current active record, 0 = Historical).

3. gold.DimCustomer

Column	Description
SK_Customer	Primary Surrogate Key identifying each unique customer
CustomerID	Original business entity identifier for the customer account.
PersonID	Individual person business entity ID (NULL if reseller).
StoreID	Reseller store business entity ID (NULL if individual).
TerritoryID	Identifier for customer's assigned sales region.
AccountNumber	Unique system account number assigned to the customer.
CustomerName	Full combined individual name or store company name.
StoreName	Retail store name or default label for individual accounts
CustomerType	Account type flag ('Individual', 'Store', or 'Store & Individual').
StartDate	Effective start date for customer record version
EndDate	Expiration date for historical customer record version.
IsCurrent	Flag indicating if customer record is currently active (1/0).
EndDate	Expiration date for historical record versions (SCD Type 2).
IsCurrent	Active status flag (1 = Current active record, 0 = Historical).

4. gold.DimGeography (Dimension Table)

Column	Description
SK_Geography	Primary Surrogate Key referencing a specific geographical address.
AddressID	Original system identifier for physical street address.
AddressLine1	Primary street address line.
AddressLine2	Secondary address detail (apt, suite, building).
City	Name of the city location.
StateProvinceID	Internal system ID for state or province.
StateProvinceCode	Standard ISO code for state or province.
StateProvinceName	Full official state or province name.
CountryRegionCode	Two-character ISO country code.
CountryRegionName	Full official name of the country.
TerritoryID	Identifier for regional sales territory.
Territoryname	Name of assigned sales territory.
TerritoryGroup	Continental or regional group for territory.

-

5. gold.DimShipMethod (Dimension Table)

Column	Description
SK_ShipMethod	Primary Surrogate Key for shipping method.
ShipMethodID	Original business identifier for shipment provider.
ShipMethodName	Commercial shipping option name.
ShipBase	Fixed base shipping charge fee.
ShipRate	Variable shipping fee rate per unit weight.

6. gold.DimCreditCard (Dimension Table)

Column	Description
SK_CreditCard	Primary Surrogate Key for credit card entry.
CreditCardID	Original business identifier for payment card.
CardType	Credit card brand type.
CardNumber	Encrypted/masked credit card account number.
ExpMonth	Variable shipping fee rate per unit weight.
ExpYear	Credit card expiration year.

7. gold.DimDate (Dimension Table)

Column	Description
DateKey	Primary integer key representing calendar date (YYYYMMDD).
Date	Standard calendar date value.
Year	Four-digit calendar year value.
Quarter	Numerical quarter of the year (1 through 4)..
Month	Numerical month index (1 through 12).
MonthName	Full text name of month.
Day	Day of the month (1 through 31).
DayOfWeek	Numerical index of day of the week (1 through 7).
DayName	Full name of weekday.

6. How to Run the Project End-to-End

Follow the steps below to execute the complete ETL pipeline:

1. **Restore the AdventureWorks2022 database** using the provided backup file (AdventureWorks2022.bak).
2. **Run initialize_db.sql**
 - Creates the Data Warehouse database.
 - Creates the Bronze, Silver, and Gold schemas.
3. **Run bronze_load.sql**
 - Extracts data from the AdventureWorks2022 source database.
 - Loads raw data into the Bronze layer without any transformations.
4. **Run bronze_profiling.sql**
 - Performs data profiling to identify duplicates, NULL values, invalid data, and referential integrity issues.
5. **Run silver_clean.sql**
 - Cleans and standardizes the Bronze data.
 - Applies business rules and creates the Silver layer.
6. **Run gold_views2_final.sql**
 - Creates the Gold layer views.
 - Builds the dimensional model (Fact and Dimension views) for analytics.
7. **Open the Power BI report**
 - Connect Power BI to the Gold layer.
 - Refresh the dataset.
 - Verify that the dashboard loads successfully.

ETL Pipeline – AdventureWorks2022

